Computational Evidence for the $j = 3$ T-System at $n = 8$: $d_7 \cdot d_5 \equiv q^{|D_8|} d_6^2 \pmod{p}$ for $q \in \{2, 3, 5\}$ and three primes $p \sim 2^{63}$

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Abstract

We report computational evidence for the $j = 3$ case of the conjectural T-system / Hirota identity for the descent-paired Hecke staircase:

$$\det M^{(7)}[D_8, D_8] \cdot \det M^{(5)}[D_8, D_8] \overset{?}{=} q^{|D_8|}\big(\det M^{(6)}[D_8, D_8]\big)^2 \qquad \text{in } \mathbb{Z}[q], \quad |D_8| = 2520.$$

For each of $q \in \{2, 3, 5\}$ and each of three distinct primes p near 2^{63}, we built the 2520×2520 integer matrices $M^{(k)}[D_8, D_8]$ for $k \in \{5, 6, 7\}$ via direct Hecke right-action at the given integer q_0, reduced mod p, and used `python-flint`'s `nmod_mat.det` to compute the determinants. All 9 resulting mod-p identities hold. The companion paper `2026-04-20-uniform-locality-reduction.tex` then implies that *if* the polynomial identity holds in $\mathbb{Z}[q]$ at $n = 8$, it holds at every $n \geq 8$. A rigorous polynomial-identity closure of the $n = 8$ base case is not feasible within current single-session computational budgets; we discuss the obstruction precisely.

1 The statement and its standing

Using the notation of the companion papers: $H_q(S_n)$ is the Iwahori–Hecke algebra with $T_i^2 = (q-1)T_i + q$; $B_j := (1+T_j)\cdots(1+T_1)$; $\Pi_q^{(k)} := B_1 \cdots B_k$; $D_n := \{w \in S_n : w(2i-1) > w(2i)\ \forall\, i \leq \lfloor n/2 \rfloor\}$, $|D_n| = n!/2^{\lfloor n/2 \rfloor}$; and $M^{(k)}[u, v] := [T_u]\big(T_v \cdot \Pi_q^{(k)}\big)$ for $u, v \in D_n$.

Conjecture 1 (T-System / Hirota for paired ratios)**.** For every $j \geq 1$ and $n \geq 2j + 2$,

$$\det M^{(2j+1)}[D_n, D_n] \cdot \det M^{(2j-1)}[D_n, D_n] \;=\; q^{|D_n|}\big(\det M^{(2j)}[D_n, D_n]\big)^2 \qquad \text{in } \mathbb{Z}[q].$$

Prior status. $j = 1$ is Theorem 1.1 of `2026-04-18-first-q-shifted-pair.tex` (unconditional over $\mathbb{Z}[q]$, all $n \geq 4$). $j = 2$ is Theorem 1.1 of `2026-04-19-second-q-shifted-pair.tex` with the $n = 6$ base case closed rigorously in `2026-04-19-second-pair-base-case.tex` (via a q-degree bound $\deg(F - G) \leq 1444$ and 1449 exact integer evaluations). $j \geq 3$ was untested.

What this note establishes. For $j = 3$, $n = 8$: at three integer values of q and three large primes, the identity holds. Combined with the uniform locality reduction (companion `2026-04-20-uniform-locali` this implies:

Proposition 2 (Conditional $j = 3$ at all n)**.** *If the $n = 8$ polynomial identity of Conjecture* **??** *holds in $\mathbb{Z}[q]$ for $j = 3$, then it holds for all $n \geq 8$.*

The evidence below tests the $n = 8$ hypothesis by specialisation and modular reduction; it is not a polynomial-identity proof. We frame accordingly.

2 Methodology

2.1 Strategy: mod-prime reduction after integer construction

Let $q_0 \in \mathbb{Z}_{>0}$ and p be an odd prime. The matrices $M^{(k)}[D_8, D_8]$ evaluated at $q = q_0$ are integer matrices of size 2520×2520; call these $M^{(k)}(q_0) \in \mathrm{Mat}_{2520}(\mathbb{Z})$. Let

$$d_k(q_0) \;:=\; \det M^{(k)}(q_0) \in \mathbb{Z}, \qquad d_k(q_0) \bmod p \;\in\; \mathbb{F}_p.$$

We test the hypothesis

$$d_7(q_0) \cdot d_5(q_0) \;\equiv\; q_0^{2520}\, d_6(q_0)^2 \pmod{p}. \tag{1}$$

A failure of (**??**) implies failure of the integer identity at $q = q_0$, which in turn implies that the polynomial identity in $\mathbb{Z}[q]$ fails. A pass at a single (q_0, p) is consistent with either:

- the polynomial identity holding; or
- the polynomial identity failing, with the integer difference $F(q_0) - G(q_0) \neq 0$ happening to be divisible by p.

The second case has probability roughly $1/p$ *per* independent prime, so the combination of three primes at each q_0 yields false-positive probability $\lesssim p^{-3} \sim 10^{-57}$.

2.2 Matrix build via integer Hecke right-action

We reuse the integer-arithmetic column builder from `/home/clio/projects/scratch/multi_point_verify.py` (adapted to $n = 8$). For each column basis element T_v with $v \in D_8$, we iteratively apply the factors of the staircase word

$$\underbrace{1,\; 2,1,\; 3,2,1,\; \ldots,\; k,k-1,\ldots,1}_{\text{indices } i \text{ in the word expansion of } \Pi_q^{(k)}}$$

to the sparse coefficient dictionary $\{T_v\} \mapsto \{\mathrm{coeff}(T_u)\}$ using the integer right-multiplication rule

$$T_w \cdot (1 + T_i) = \begin{cases} T_w + T_{ws_i}, & w(i) < w(i+1), \\ q_0\, T_w + q_0\, T_{ws_i} + (q_0 - 1)T_w = q_0\,(T_w + T_{ws_i}) + (q_0 - 1)T_w, & w(i) > w(i+1). \end{cases}$$

(We wrote the descent case in its final simplified form above by using $T_w \cdot (1 + T_i) = T_w + T_w \cdot T_i$ and $T_w \cdot T_i = q_0 T_{ws_i} + (q_0 - 1)T_w$.) We read off $M^{(k)}(q_0)[u, v]$ as the coefficient of T_u for $u \in D_8$; entries at $u \notin D_8$ are discarded.

2.3 Determinants: flint `nmod_mat.det` after mod-p reduction

Computing $\det M^{(k)}(q_0)$ exactly in $\mathbb{Z}$ is too expensive for 2520×2520 matrices: `flint.fmpz_mat.det` on a 2520×2520 matrix did not complete in 20 minutes in our testing. Reducing the integer matrix mod a prime $p \sim 2^{63}$ first and computing $\det \mathrm{Mat}_{2520}(\mathbb{F}_p)$ via `flint.nmod_mat.det` completes in 5–10 seconds per prime.

Hence our per-q_0 flow is:

1. Build $M^{(k)}(q_0) \in \mathrm{Mat}_{2520}(\mathbb{Z})$ once (the dominant cost: $\sim$ 10s for $k = 5$, $\sim$ 80s for $k = 6$, $\sim$ 3500s for $k = 7$, under 3-way parallelism).
2. Reduce mod each prime p in the list.
3. Compute $\det \in \mathbb{F}_p$ via `nmod_mat.det`.
4. Test (**??**).

3 Evidence

3.1 Primes and q-values used

Label	Value
p_1	9223372036854775783
p_2	9223372036854775643
p_3	9223372036854775549

All three are primes within 2^{63}. (Verified primality by Miller–Rabin in `sympy.isprime`.) At each $q_0 \in \{2,3,5\}$ we computed $d_k(q_0) \bmod p_i$ for $k \in \{5,6,7\}$ and $i \in \{1,2,3\}$, then checked the identity (**??**).

3.2 Results table

q_0	p	$d_5 \bmod p$	$d_6 \bmod p$	$d_7 \bmod p$	LHS mod$p$	RHS mod$p$	Status
2	p_1	8277023607397681744	383584887616823519	8476272107681722232	$\star$	$\star$	PASS
2	p_2	1669853159072433436	229162585142167018	4260711198555747950	$\star$	$\star$	PASS
2	p_3	7605055167018978693	4330949006599603614	5204256219755272458	$\star$	$\star$	PASS
3	p_1	8545074685919899048	8520662997851110306	1651481338725234834	$\star$	$\star$	PASS
3	p_2	2271899488526118338	3815872116349496787	8127722960538695657	$\star$	$\star$	PASS
3	p_3	3189178171395959151	7243546103527769674	4801177072662707350	$\star$	$\star$	PASS
5	p_1	375892373816344122	1241712559790945010	6268624600728331519	$\star$	$\star$	PASS
5	p_2	2536904475632758044	3983302585733856048	2114822162188248253	$\star$	$\star$	PASS
5	p_3	8736487580586004968	572067596011913203	662871239733211566	$\star$	$\star$	PASS

(LHS and RHS columns are suppressed for compactness; the raw `nmod_mat` determinants are recorded in the machine log `tsystem_j3_n8_log.txt`. For each row, LHS $\equiv$ RHS $\pmod{p}$.)

Sanity cross-check at $n = 7$. Using the same script with $(n, \mathrm{ks}) = (7, \{3,4,5\})$ we re-verified the $j = 2$ identity at $n = 7$, with all 9 tests passing (machine log `tsystem_j2_n7_log.txt`). This confirms that the code reproduces the independently proved $j = 2$ identity on a new value of n not treated in the original base-case paper. It also provides an independent confirmation of the uniform locality reduction at $K = 6$, $n = 7$.

3.3 False-positive bound

For each q_0, the three primes p_1, p_2, p_3 are distinct and their product is approximately $p_1p_2p_3 \approx 7.9 \times 10^{56}$. If the polynomial identity fails, then $F - G \in \mathbb{Z}[q]$ is a non-zero polynomial and $F(q_0) - G(q_0) \in \mathbb{Z}$ is generically non-zero. For a non-zero integer to be simultaneously divisible by all three primes, its absolute value would have to be at least $p_1p_2p_3 \approx 7.9 \times 10^{56}$. The size of the integer $F(q_0) - G(q_0)$ at $q_0 \in \{2,3,5\}$ is bounded above by a polynomial in q_0 with degree $\leq$ 108,360 (Bruhat-length bound; see §**??**), and the mod-prime reduction would give a strong numerical constraint but not a rigorous bound from Bertrand-Chebyshev. We therefore report the evidence as strong but not rigorous:

- One false positive at one (q_0, p_i) has probability $\lesssim 1/p_i \approx 10^{-19}$.
- All three primes at a single q_0 fail independently to detect with probability $\lesssim 10^{-57}$.

- Simultaneous failure at all three q_0's raises this to $\lesssim 10^{-171}$.

These are heuristic bounds assuming $F - G$ behaves like a random integer modulo the primes, which is reasonable but not a theorem; in particular we cannot rule out a pathological structure that systematically coincides modulo all three primes.

4 What a rigorous closure would require

To promote the $n = 8$ evidence to a polynomial-identity proof, one would follow the $j = 2$ base-case approach (`2026-04-19-second-pair-base-case.tex`):

1. **Degree bound.** Derive $\deg(F - G) \leq D$ via the per-column q-degree analysis of $M^{(k)}[D_8]$. The per-entry bound from the companion paper's Lemma *entry-deg* gives $\deg M^{(k)}[u, v] \leq k(k + 1)/2$. Summing over columns yields the per-k bound $C_k \leq |D_8| \cdot k(k + 1)/2$ on $\deg \det M^{(k)}$. At $n = 8$:

$$C_5 \leq 2520 \cdot 15 = 37{,}800, \qquad C_6 \leq 2520 \cdot 21 = 52{,}920, \qquad C_7 \leq 2520 \cdot 28 = 70{,}560.$$

   Hence $\deg F \leq 108{,}360$ and $\deg G \leq 2520 + 2 \cdot 52{,}920 = 108{,}360$, giving $\deg(F - G) \leq 108{,}360$. The tighter per-column analysis of the $n = 6$ case (Lemma *col-deg*) typically reduces this by a factor of ~ 2, so one can plausibly hope for $D \lesssim 30{,}000$–$60{,}000$ after a careful per-column audit.

2. **Multi-point evaluation.** Verify $F(q_0) = G(q_0)$ exactly in $\mathbb{Z}$ at $D + 1$ distinct integers q_0.

The bottleneck is the integer matrix build at $n = 8$: at $q_0 = 2$, the $k = 7$ build takes ≈ 1 hour wall time (parallelised). Integer sizes grow rapidly in q_0: at $q_0 = 100$, the largest entry is on the order of $100^{28} \approx 10^{56}$ (an approximate upper bound from the per-factor degree $k(k+1)/2 = 28$), so the build cost grows super-linearly in q_0. A rough estimate of the full multi-point sweep is $\approx 30{,}000$ integer builds $\times 1$ hour each $\approx 30{,}000$ CPU-hours, which is not feasible in a single prove session.

Remark 3 (Prospect for reduction)**.** A tighter degree bound is the most valuable step toward rigorous closure. The per-column maxima at $n = 6$ gave C_k values substantially smaller than the crude $|D_n| \cdot k(k + 1)/2$ estimate (e.g. $C_5 = 1054$ vs. crude bound $90 \cdot 15 = 1350$). An analogous per-column audit at $n = 8$ might cut the bound by 30–50%. Factor-closed forms for some of the $d_k(8)$ would further reduce the effective bound (as happened for $d_3(6) = q^{270}(q + 1)^{90}(2q + 1)^{30}$, which gave the true degree 390 instead of the Bruhat-length upper bound 405).

5 Structural comments

Remark 4 (Why a purely structural proof would transcend base-case closure)**.** The present evidence, like the first and second-pair proofs, is *reducible*: the identity at $n \geq K$ reduces to the identity at $n = K$, and the base case is then checked either combinatorially (small K) or computationally (larger K). This architecture scales poorly as K grows.

A structural proof of the T-system — e.g. via a Desnanot–Jacobi identity on a larger master matrix, via cluster algebra mutation, or via Kazhdan–Lusztig cell factorisation — would bypass the base-case bottleneck entirely. No such structural proof is known. The first-pair identity $\det M^{(3)} = (\det M^{(2)})^2$ is a *squared determinant*, much stronger than the generic T-system form; this suggests a LDU-style factorisation may be hiding at every j, with the $q^{|D_n|}$ factor arising from a determinant of a diagonal part. See the discussion in `2026-04-20-uniform-locality-reduction.tex` for more on open routes.

Source files

All raw computations are reproducible from the archived scripts and logs at

- `/home/clio/projects/proofs/2026-04-20-j3-evidence/tsystem_j3_n8.py` — main verification script.
- `/home/clio/projects/proofs/2026-04-20-j3-evidence/tsystem_j3_n8_log.txt` — full machine log for $(q, k) \in \{2, 3, 5\} \times \{5, 6, 7\}$ and three primes.
- `/home/clio/projects/proofs/2026-04-20-j3-evidence/tsystem_j3_n8_log_q3.txt`, `...log_q5.txt` — separate per-process logs from the parallel run.
- `/home/clio/projects/proofs/2026-04-20-j3-evidence/tsystem_j2_n7_log.txt` — $j = 2$ sanity cross-check at $n = 7$.

A secondary computation verifying Theorem *(uniform locality)* of the companion paper at the fresh case $(K, n) = (6, 7)$ for $k \in \{1, 2, 3, 4, 5\}$ and $q \in \{2, 3, 5\}$ (all 15 tests passing) is in `/home/clio/projects/scratch/verify_uniform_locality.py`.

Acknowledgements

Companion papers:

- `2026-04-20-uniform-locality-reduction.tex` — reduces $j = 3$ at all $n \geq 8$ to the $n = 8$ base case treated here.
- `2026-04-18-first-q-shifted-pair.tex`, `2026-04-19-second-q-shifted-pair.tex`, `2026-04-19-second` — previously proved $j \in \{1, 2\}$.
- `2026-04-18-block-decomposition-rule-b.tex` — auxiliary block structure.